Abhi Tyagi

Dublin, Ireland

+353-830134217 abhi.tyagi994@gmail.com linkedin.com/in/abhi-tyagi github.com/abhityagi994-prog portfolio

Professional Summary

MSc Business Analytics graduate (First Class Honours) with 3+ years across Python, SQL, and end-to-end machine learning delivery, including 2 years on a 500,000+ user US healthcare platform at Wipro-Alight. Built and deployed ML systems covering NLP, recommendation, and time-series forecasting, with reproducible pipelines using Docker, MLflow, DVC, and GitHub Actions CI.

Technical Skills

Languages: Python, SQL
Machine Learning: Supervised & Unsupervised Learning, Deep Learning, NLP, Recommendation Systems, Time-Series Forecasting, Model Evaluation, Explainable AI, Predictive & Prescriptive Modelling
Generative AI: LLMs, RAG, Prompt Engineering
Frameworks & Libraries: Scikit-learn, TensorFlow, Pandas, NumPy
Applications & Visualization: Flask, Streamlit, Power BI, DAX, Matplotlib, Plotly
Software Practices: Git, GitHub Actions, Pytest, Modular Coding, Documentation, CI/CD
MLOps & Cloud (AWS): MLflow, DVC, Docker, S3, ECR, ECS/Fargate, EC2, IAM, CloudWatch
Databases & Tools: MySQL, Jupyter, REST APIs

Projects

YouTube Comment Sentiment Analysis
Python, LightGBM, NLTK, TF-IDF, Flask, MLflow, DVC, Docker, GitHub Actions, AWS (ECR, ECS/Fargate, S3, CloudWatch)

- Built a three-class sentiment classification system on YouTube comments using TF-IDF vectorisation and LightGBM, exposed via a Flask REST API and integrated into a Chrome extension for real-time inference on live videos.
- Implemented reproducible ML pipeline using DVC and MLflow model registry with candidate to champion alias promotion, gated by GitHub Actions CI running automated model loading, signature, and performance validation tests.
- Containerised the inference service with Docker, published to Amazon ECR, and deployed on Amazon ECS/Fargate with CloudWatch logging and IAM task roles for secure runtime AWS access.

Experience

Data Scientist **May 2025 – Present**
Freelance *Remote*

- Delivered a hybrid music recommendation engine for a client building a discovery product, combining content-based similarity with collaborative filtering and an automatic cold-start fallback, shipped as a Dockerised Streamlit app on Amazon EC2 pulling private images from Amazon ECR.
- Built a DVC-tracked reproducible pipeline across cleaning, collaborative filtering, and hybrid transformation stages, with artifacts versioned on Amazon S3 and GitHub Actions CI validating tests on every change, so the client could retrain and redeploy independently.
- Developed equity movement forecasting models on 73,608 Reddit articles joined with Yahoo Finance time-series data, benchmarking Gradient Boosting, LSTM, CNN, and CNN-LSTM to reach 56.1% accuracy and 57.1% ROC-AUC on S&P 500 directional prediction.
- Applied SHAP explainability to surface model drivers and walked technical and non-technical stakeholders through model behaviour before sign-off.

Data Analyst **April 2022 – March 2024**
Wipro Limited — Client: Alight Solutions *Noida, India*

- Partnered with US-based stakeholders on a 500,000+ user healthcare platform, translating open-ended business challenges into structured analytical questions and solution plans using SQL and Python.
- Developed Python code for data collection, cleaning, transformation, and reusable analytical workflows, identifying trends, anomalies, and data quality issues across structured and unstructured datasets.
- Built automated reporting workflows and KPI dashboards tracking pass rates, completion rates, and defect trends across quarterly release cycles, reducing manual effort by approximately 100 hours per quarter.
- Delivered analytical insights across 12–13 concurrent projects per release cycle, taking ownership of accuracy and communicating recommendations clearly to both technical and non-technical stakeholders.

Certifications

Oracle Certified Generative AI Professional • Oracle Certified OCI AI Foundations Associate • Data Science Mentorship Program (DSMP 2.0) — CampusX

Education

Dublin Business School **Dublin, Ireland**
MSc in Business Analytics, First Class Honours *April 2024 – April 2025*

Meerut Institute of Engineering and Technology (MIET) **India**
B.Tech in Mechanical Engineering, First Division with Distinction *August 2018 – July 2022*